Time series Analysis - MATH1318

Assignment_1

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2026-03-25

###INTRODUCTION: ####In this analysis, I examine a time series dataset of Victorian energy demand, where the observations are recorded in a sequence with time and are dependent on time. In contrast to independent data, time series data is affected by its previous values, and it is significant to examine both deterministic and random elements of the series. The data contains 152 observations in GW/100 units, which I model as a univariate time series to gain insight into the underlying patterns and behaviour.

####My primary goal is to determine the trend and cyclical patterns of the energy demand series using time series models based on regression. To do this, I use four various deterministic models: a linear trend model, a quadratic trend model, a cyclical model, and a cosine-based model. These models are used to capture the systematic structure in the data, and the random variation is analysed using residual diagnostics. I compare the performance of both models based on statistical values like Adjusted R 2, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and Root Mean Square Error (RMSE).

####I also conduct diagnostic checking to determine the satisfaction of the model assumptions, such as independence, constant variance, and normality of residuals. It is done with the help of residual plots, histograms, QQ plots, and autocorrelation function (ACF) plots. I choose the most effective model to predict the future values of energy demand based on the comparison of all models. Short-term forecasts and prediction intervals are then produced using the chosen model, which gives information about the future behaviour of the series.

```r
library(tinytex)
rm(list = ls())
```

###Data Import and Preliminary Inspection: #### In this step, first, the necessary package "TSA" (Time Series Analysis) is loaded. The TSA package includes a range of important functions required to analyse time series data. These functions include modelling, diagnostics, and statistical functions.

####Then, the dataset "assignment1Data2026.csv" is imported into R using the "read.csv()" function. The working directory is also set to access the required file. The dataset includes two variables, "Minute" indicating the time index, and "Demand (GW/100)" indicating the energy demand values. This dataset is used to perform time series analysis, where the demand is recorded over time.

```r
# Loading the TSA package for time series analysis functions
library(TSA)
```

```
##
## Attaching package: 'TSA'

## The following objects are masked from 'package:stats':
##
##     acf, arima
```

```
## The following object is masked from 'package:utils':
##
##     tar
```

```r
# Reading the data
setwd("~/Downloads/timeseries_assignment")
energy <- read.csv("assignment1Data2026.csv", header = TRUE)
```

###Time Series Construction: ####In this step, I transform the energy demand data into a formal time series object using the ts() function. In particular, I select the demand data and create a time series object with a single variable starting at time t = 1 with a frequency of 1. This implies that the data is recorded at equally spaced time intervals with no indication of a seasonal frequency.

####By transforming the data into a time series object (demandTS), I will now able to take advantage of various time series functions that may need for further analysis. The time series object consists of 152 observations representing the demand for energy over time.

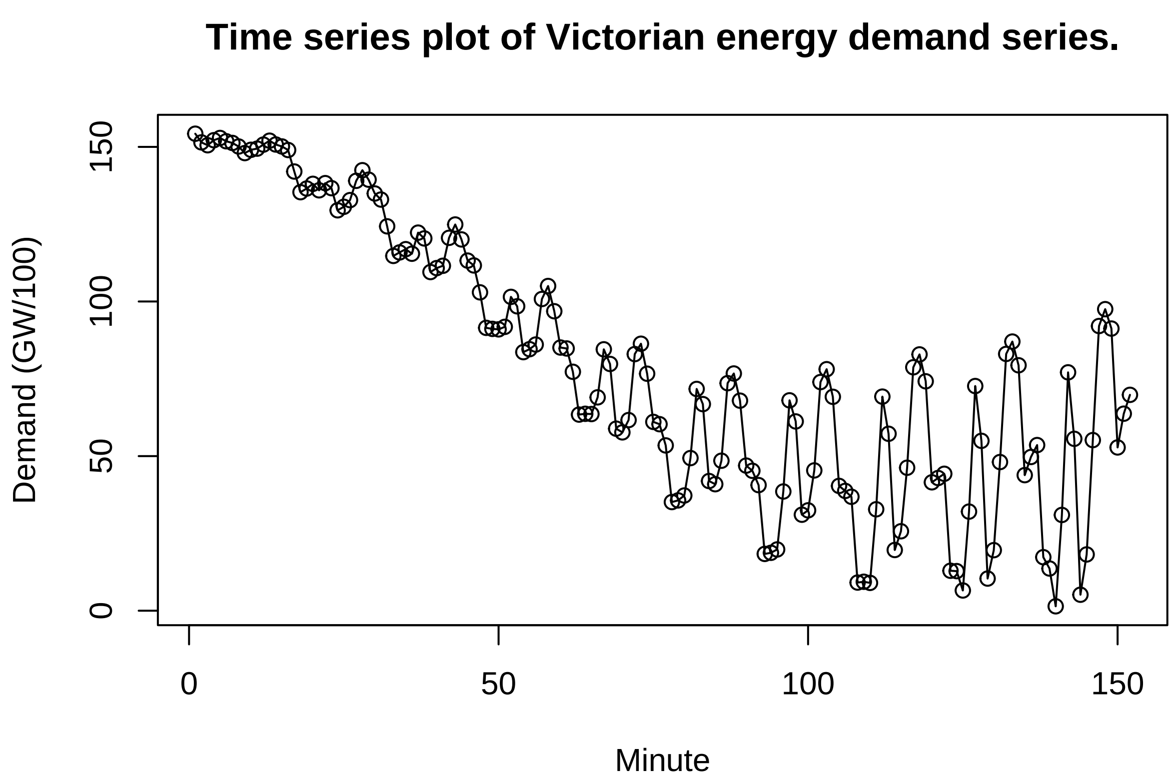
```r
# Creating time series object
demandTS <- ts(energy$Demand..GW.100., start = 1, frequency = 1)
demandTS
```

```
## Time Series:
## Start = 1
## End = 152
## Frequency = 1
##   [1] 154.262289 151.455117 150.535458 152.162375 152.883087 151.819494
##   [7] 151.273992 150.128478 147.998420 149.058493 149.435896 150.736413
##  [13] 152.008609 150.773967 150.133731 148.989800 142.037106 135.352146
##  [19] 136.539042 138.057735 135.960266 138.279890 136.656488 129.492703
##  [25] 130.627754 132.785314 139.030234 142.460441 139.401278 134.935186
##  [31] 132.967459 124.309947 114.738740 115.867658 116.917242 115.451659
##  [37] 122.275495 120.398128 109.525456 110.806623 111.549865 120.591477
##  [43] 124.903861 120.077037 113.220435 111.664627 102.956479  91.483398
##  [49]  91.142408  90.986536  91.798966 101.494565  98.442501  83.628734
##  [55]  84.578581  86.072398 100.805176 105.013420  96.846911  85.114707
##  [61]  84.810229  77.266505  63.429548  63.687294  63.600691  68.998837
##  [67]  84.533837  79.802596  58.879580  57.674999  61.655129  82.987951
##  [73]  86.343179  76.647931  61.061601  60.321755  53.456340  35.140417
##  [79]  35.677030  37.258740  49.362270  71.718446  66.813185  41.939410
##  [85]  40.913505  48.504052  73.584742  76.712331  67.936990  46.927101
##  [91]  45.253109  40.625217  18.372914  18.731776  19.811776  38.553330
##  [97]  68.041715  61.215626  31.043157  32.460221  45.366804  73.939817
## [103]  78.102172  69.179439  40.364495  38.739480  36.847932   9.103143
## [109]   9.361855   9.000031  32.793716  69.237751  57.220211  19.632194
## [115]  25.691440  46.243740  78.721590  82.889599  74.190155  41.525242
## [121]  42.915826  44.283743  12.922689  12.805951   6.546039  32.052143
## [127]  72.659149  54.941327  10.400125  19.582447  48.080005  83.046583
## [133]  87.036962  79.373471  43.842551  49.711307  53.565955  17.311508
## [139]  13.644374   1.428485  30.969435  77.086788  55.594202   5.172394
## [145]  18.193651  55.178218  92.065289  97.524512  91.243905  52.803228
## [151]  63.721560  69.817186
```

####In this step, I constructed a time series plot of the Victorian energy demand data to visually examine its underlying structure over time. The time index t is generated using the time() function, which is later used as an explanatory variable in regression-based modelling. The plot displays the demand values against time, allowing me to identify key characteristics such as trend,seasonality,changing varience,behaviour and change point in the series.

####Trend: Upward quadratic trend, where the series initially decreases and then gradually increases over time, indicating a non-linear trend structure. ####Seasonality: No seasonality is observed, as there is no consistent repeating pattern. ####Behavior: Auto regressive behaviour is observed, as consecutive observations are highly dependent on their previous values, indicating strong temporal dependence. ####Change Point: No change point is identified, as the trend changes smoothly without any abrupt structural shifts. ####Changing Variance: No changing variance is observed, the variance remains relatively constant over time, indicating no significant heteroscedasticity.

```r
# Creating time points
t <- time(demandTS)

# Time series plot
plot(demandTS, ylab = "Demand (GW/100)", xlab = "Minute", type = "o",
     main = "Time series plot of Victorian energy demand series.")
```



####In this step, a new variable y is created and the time series object demandTS is assigned to it. This is simply a matter of simplifying the notation and making the data ready for further use, especially when comparing current and lagged values as will be required later. The head(y) function is then used to display the first few values of the series, enabling me to quickly look at the data and confirm its storage.

```r
# Extraction and preview of current time series values
y <- demandTS
head(y)
```

```
## [1] 154.2623 151.4551 150.5355 152.1624 152.8831 151.8195
```

####In this step, the lagged version of the given time series is formed by utilizing the zlag() function. This function shifts the given series by a unit of one, where every value in the series x represents the previous value in the demand series.

####The first value in this series is NA, as there is no previous value for the first value in the series. The head(x) function is used to view the initial values in this lagged series, which could be used for verifying whether the lag operation has been correctly performed or not.

```r
# Creating lagged values of the demand series
x <- zlag(demandTS)
head(x)
```

```
## [1]       NA 154.2623 151.4551 150.5355 152.1624 152.8831
```

###Correlation Analysis of Current and Lagged Values:

####The cor() function is used to determine the correlation coefficient by Pearson method of assessing the strength and direction of linear relationships between consecutive values of y (current values) and x (lagged values).

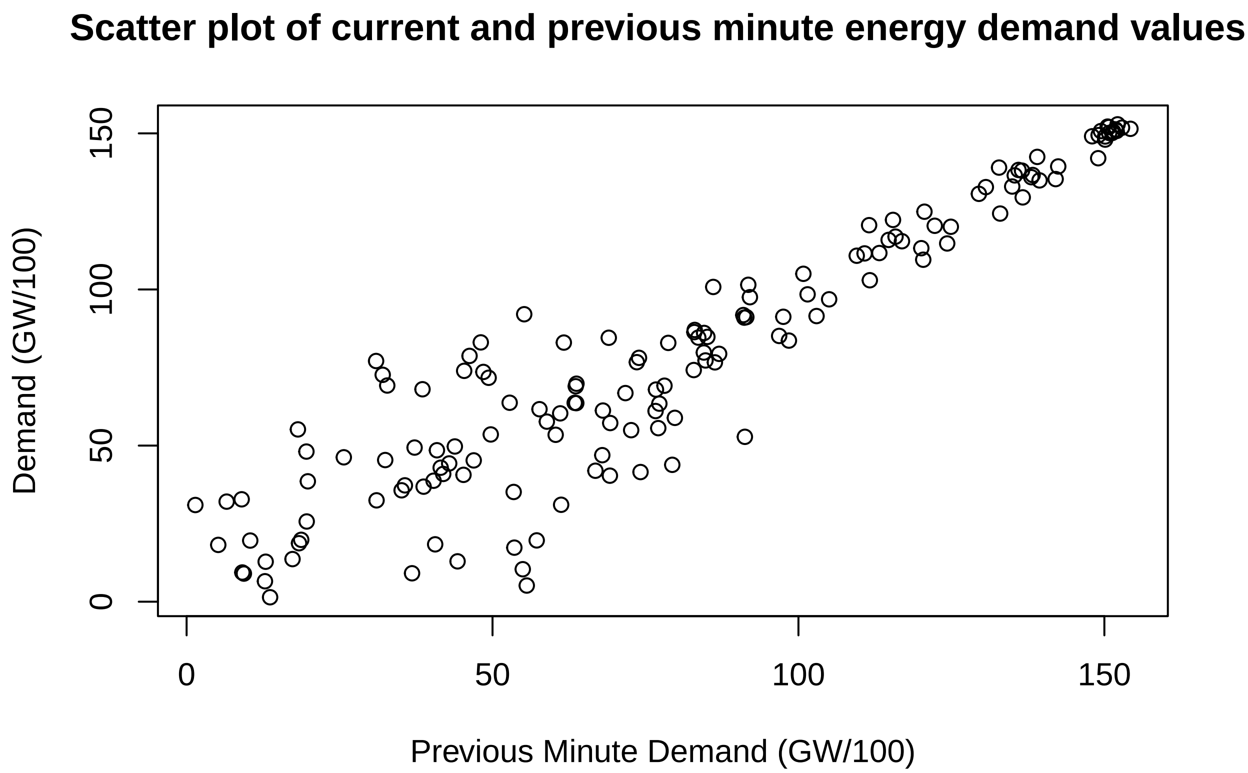
####The correlation coefficient is obtained and it is 0.9305, indicating a very strong positive linear relationship. This implies that the present value of demand is strongly dependent on the past value of demand, and it is strongly autocorrelated, which is generally apparent in autoregressive time series.

```r
# Calculating correlation between current and lagged demand values
index <- 2:length(x)
cor(y[index], x[index])
```

```
## [1] 0.9305432
```

####Based on the scatter plot, The plot shows that there is an upward linear trend, and the points are in a definite positive linear trend. This implies that the higher the past minute demand, the higher the present demand, and there is a strong positive correlation between the successive observations.

```r
# A scatter plot showing the correlation between minute demand values from past and present.
plot(y = y[index], x = x[index], ylab = "Demand (GW/100)",
     xlab = "Previous Minute Demand (GW/100)",
     main = "Scatter plot of current and previous minute energy demand values.")
```



###Model1:Linear model ####This step involves the fitting of a linear regression model to the time series data based on the model demandTS = beta 0 + beta 1 t + epsilon t, where t is time. This model represents the general linear trend in the energy demand series.

####Based on the model output,The results indicate that the intercept is estimated to be 142.7864, which is the level of demand at t = 0. The slope coefficient is -0.8168, which shows a negative linear trend, with the demand declining with time. The p-values of the two coefficients are below alpha (p-value < 2e-16) at a significance level of alpha = 0.05, which proves that the coefficients are statistically significant and time is a significant predictor of demand.

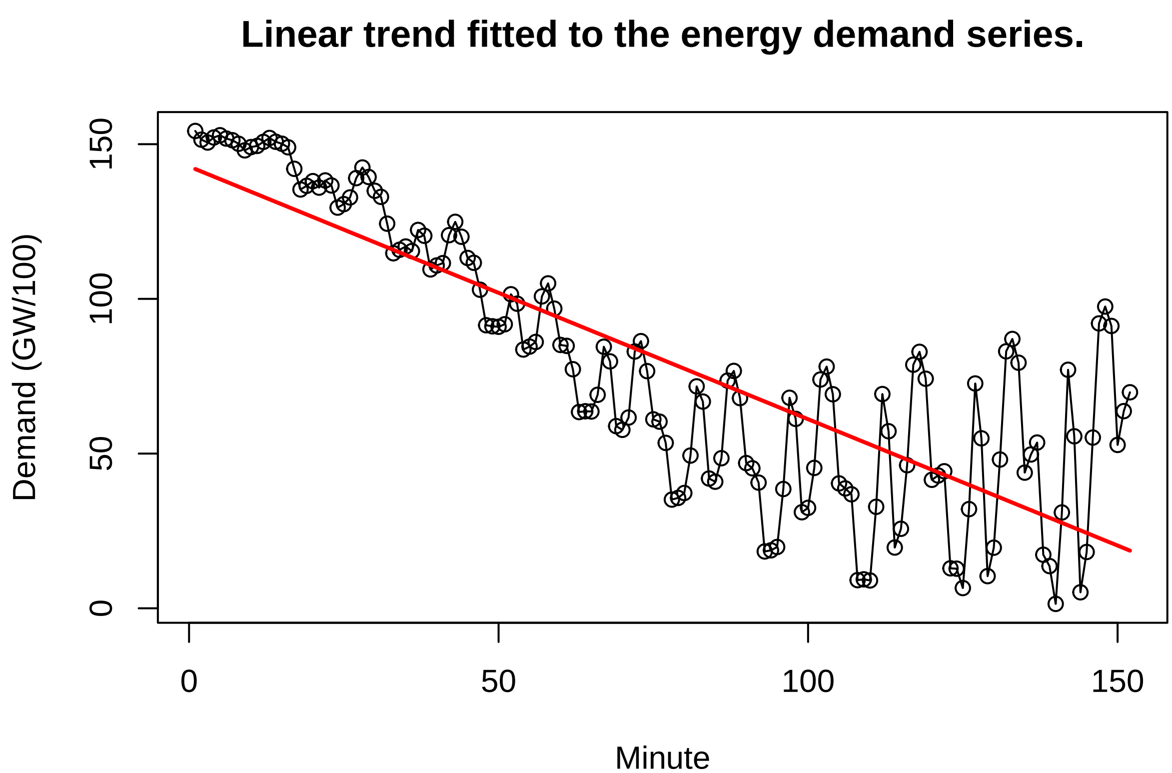
####I also note that the model accounts a significant percentage of the variability in the data with an Adjusted R 2 of 0.6845, which means that about 68.45 percent of the variation in demand is accounted by the linear trend. The Residual Standard Error (24.38) is the mean difference between the observed values and the fitted line.

```r
# Model specification and fitting
# Model 1: Linear trend model
model1 <- lm(demandTS ~ t)
summary(model1)
```

Based on the plot, It is observed that the linear model is able to explain the general downward trend in the series, but it does not explain the variation and curvature in the data. This suggests a more complicated model may be necessary to capture the underlying structure of the series.

```
##
## Call:
## lm(formula = demandTS ~ t)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -48.455 -18.743   1.518  13.890  75.618
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 142.78643    3.97385   35.93   <2e-16 ***
## t            -0.81675    0.04506  -18.13   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 24.38 on 150 degrees of freedom
## Multiple R-squared:  0.6866, Adjusted R-squared:  0.6845
## F-statistic: 328.5 on 1 and 150 DF,  p-value: < 2.2e-16
```

```
plot(demandTS, ylab = "Demand (GW/100)", xlab = "Minute", type = "o",
     main = "Linear trend fitted to the energy demand series.")
lines(ts(fitted(model1), start = 1, frequency = 1), col = "red", lwd = 2)
```



###Model2:Quadratic Model

####Here, the linear model is further developed by adding a quadratic t 2 to the linear demand equation. This provides a new equation: demandTS = beta_0 + beta_1 t + beta_2 t 2 + epsilon_t.

####According to the result, the intercept value is approximated to be 177.0 units, which is the level of the demand at the base. The coefficient of t is estimated to be -2.151 and the coefficient of t 2 is also estimated to be 0.008722 which would mean that demand changes non-linearly, where it goes down first and then upwards as time goes by.

####According to the produced output, the p-value of each term at alpha = 0.05 is lesser than alpha (p-value = less than 2e -16) indicating that all the terms of the equation are statistically significant.

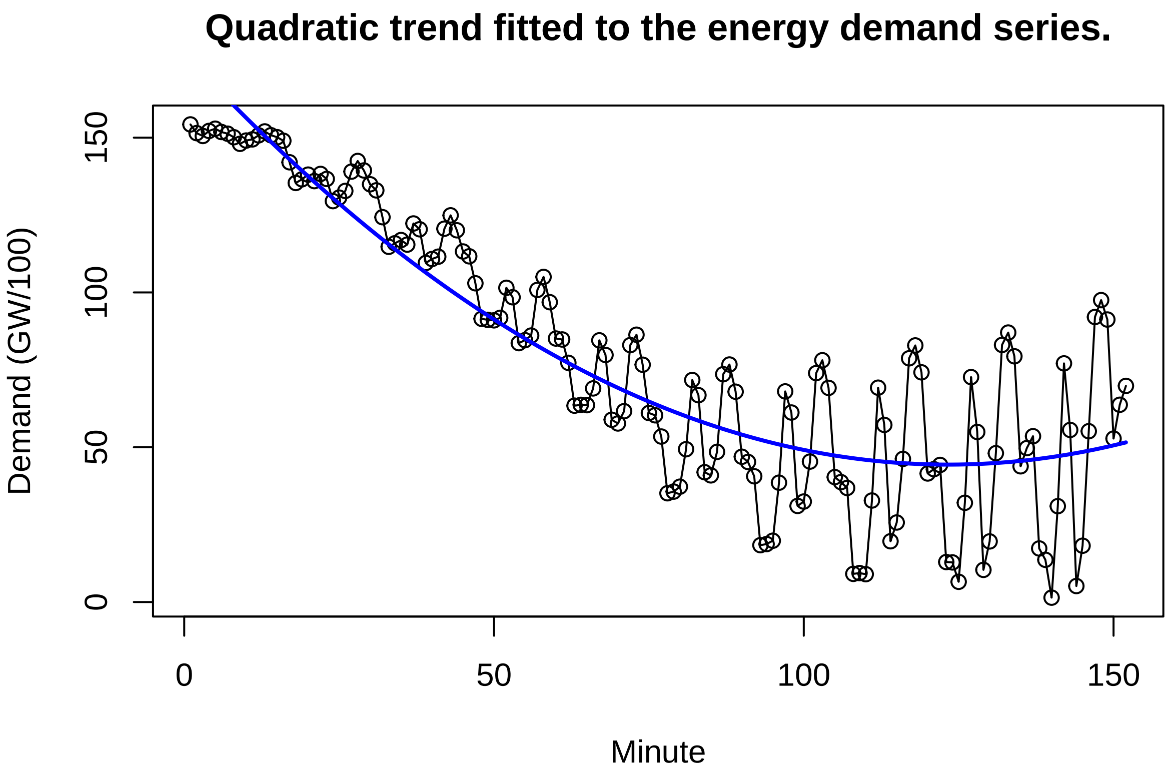
####With the output, the new model outperforms the linear model since the Adjusted R-squared has been estimated to be 0.8045, which means that on the demand, the model has been found to explain about 80.45% variability. The Residual Standard error is also lesser than that of the linear model (19.19).

####Based on the plot, the quadratic curve (blue line) aligns more with the data in comparison with the linear curve that does not match the data pattern of decline and consequent increase in demand. This implies that the quadratic model is much closer to the data as compared to the linear model.

```r
# Model 2: Quadratic trend model
t2=t^2
model2 <- lm(demandTS ~ t + t2)
summary(model2)
```

```
##
## Call:
## lm(formula = demandTS ~ t + t2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -45.390 -11.427   0.404  12.546  47.821
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.770e+02  4.730e+00  37.425   <2e-16 ***
## t           -2.151e+00  1.427e-01 -15.070   <2e-16 ***
## t2           8.722e-03  9.037e-04   9.651   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.19 on 149 degrees of freedom
## Multiple R-squared:  0.8071, Adjusted R-squared:  0.8045
## F-statistic: 311.8 on 2 and 149 DF,  p-value: < 2.2e-16
```

```r
plot(demandTS, ylab = "Demand (GW/100)", xlab = "Minute", type = "o",
     main = "Quadratic trend fitted to the energy demand series.")
lines(ts(fitted(model2), start = 1, frequency = 1), col = "blue", lwd = 2)
```



Model3:Cyclical Model

####In this step, I am expanding the quadratic model by incorporating a cubic term, i.e., t^3, which gives rise to the model demandTS = beta_0 + beta_1 t + beta_2 t^2 + beta_3 t^3 + epsilon_t This model also offers greater flexibility in capturing complex patterns in the given time series.

####As can be observed from the above model output, the intercept term in the model gets estimated as 159.0, and the coefficient estimates for t, t^2, and t^3 are statistically significant. As the significance level alpha = 0.05, the p-values for these coefficient estimates are less than alpha, which confirms that these terms are contributing significantly to the model, as observed in the above model output.

####the model performance also gets improved in this case, as indicated by the high value of Adjusted R-Squared, i.e., 0.8267, which suggests that approximately 82.67% of the demand variations are explained in this case, with a lower value of the Residual Standard Error, i.e., 18.07, as compared to the linear and quadratic models.

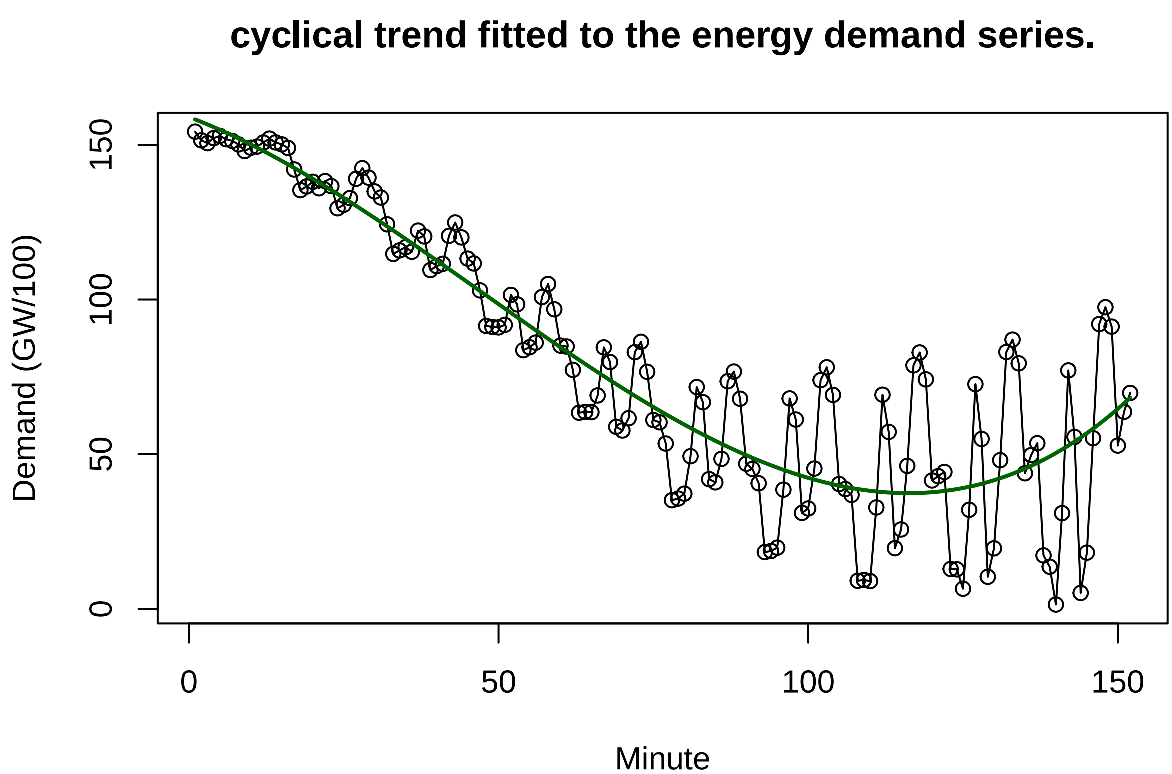
####From the plot, I note that the fitted curve, as indicated by the green line, seems to closely track the variations in the series, especially in the decrease and then rise, which are more closely modeled by the cubic curve. This indicates a better fit for the cubic model in representing the underlying structure of the series compared to other models.

```
# Model 3: cyclical model
# A polynomial in time can be used as a simple cyclical representation
model3 <- lm(demandTS ~ t + I(t^2) + I(t^3))
summary(model3)
```

```
##
```

```
## Call:
## lm(formula = demandTS ~ t + I(t^2) + I(t^3))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -50.228  -7.798  -1.623   8.634  45.401
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.590e+02  6.009e+00  26.458  < 2e-16 ***
## t           -7.583e-01  3.390e-01  -2.237  0.02679 *
## I(t^2)      -1.396e-02  5.140e-03  -2.717  0.00738 **
## I(t^3)       9.884e-05  2.209e-05   4.475 1.51e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.07 on 148 degrees of freedom
## Multiple R-squared:  0.8301, Adjusted R-squared:  0.8267
## F-statistic: 241.1 on 3 and 148 DF,  p-value: < 2.2e-16
```

```r
plot(demandTS, ylab = "Demand (GW/100)", xlab = "Minute", type = "o",
     main = "cyclical trend fitted to the energy demand series.")
lines(ts(fitted(model3), start = 1, frequency = 1), col = "darkgreen", lwd = 2)
```



###Model4:Cosine Wave Model #### Here, I use a cosine wave model to model cyclical behaviour in the time series with sine and cosine functions. A set of potential frequencies is created and a regression model is estimated at each frequency. The best model is the one with the lowest total of squared errors (SSE) and the frequency is determined to be 0.00658.

####With this optimum frequency, the cosine and sine terms are built and added to the regression model. ####demandTS = beta_0 + beta_1 cos(2pi f t) + beta_2 sin(2pi f t) + epsilon_t.

####Based on the model output, I can see that the intercept is 80.305, and the coefficients of the cosine and sine terms are 20.246 and 45.533, respectively. The p-values of all the coefficients are below alpha at a significance level of alpha = 0.05, which means that the cyclical components are statistically significant.

####Nevertheless, the model describes a comparatively smaller percentage of variability than the earlier models, with an Adjusted R 2 of 0.6592. The Residual Standard Error (25.33) is also greater than the quadratic and cubic models, indicating a poorer fit.

####Based on the plot, I can see that the cosine curve (purple line) is able to capture a smooth oscillatory pattern, but it does not fit the actual structure of the data, especially the non-linear trend. This means that the cosine model is not as effective in capturing the underlying behaviour of the time series.

```
# Model 4: Cosine wave model

freqGrid <- seq(1/152, 20/152, length.out = 500)

sseValues <- numeric(length(freqGrid))
modelList <- vector("list", length(freqGrid))

for(i in 1:length(freqGrid))
{
  cosTerm <- cos(2*pi*freqGrid[i]*t)
  sinTerm <- sin(2*pi*freqGrid[i]*t)
  fit <- lm(demandTS ~ cosTerm + sinTerm)
  sseValues[i] <- sum(residuals(fit)^2)
  modelList[[i]] <- fit
}

bestIndex <- which.min(sseValues)
bestFreq <- freqGrid[bestIndex]
bestFreq
```

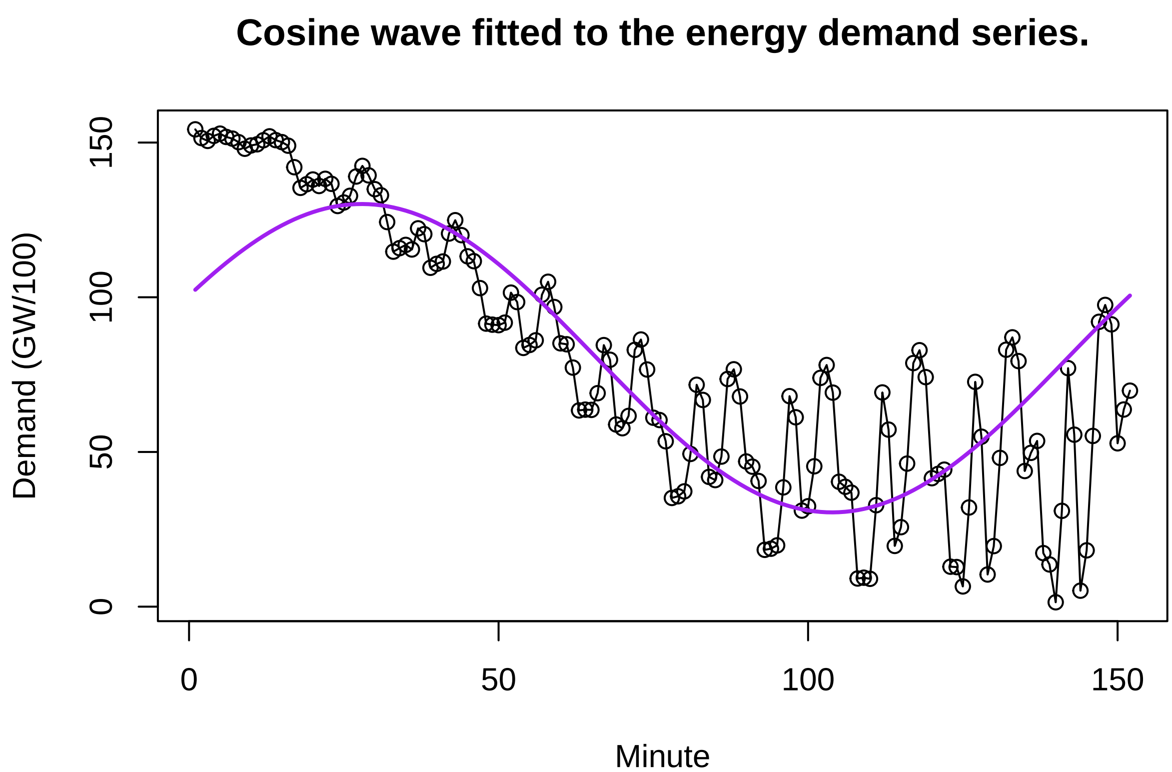
```
## [1] 0.006578947
```

```
cosTerm <- cos(2*pi*bestFreq*t)
sinTerm <- sin(2*pi*bestFreq*t)

model4 <- lm(demandTS ~ cosTerm + sinTerm)
summary(model4)
```

```
##
## Call:
## lm(formula = demandTS ~ cosTerm + sinTerm)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -79.497 -15.214   0.327  14.659  51.847
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   80.305      2.055  39.082  < 2e-16 ***
## cosTerm       20.246      2.906   6.967 9.69e-11 ***
## sinTerm       45.533      2.906  15.669  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 25.33 on 149 degrees of freedom
## Multiple R-squared:  0.6637, Adjusted R-squared:  0.6592
## F-statistic:   147 on 2 and 149 DF,  p-value: < 2.2e-16
```

```r
plot(demandTS, ylab = "Demand (GW/100)", xlab = "Minute", type = "o",
     main = "Cosine wave fitted to the energy demand series.")
lines(ts(fitted(model4), start = 1, frequency = 1), col = "purple", lwd = 2)
```



###Model Comparison: ####This step involves the development of a table to compare the models to compare the performance of all the four models: Linear, Quadratic, Cyclical, and Cosine. Four primary statistical indicators are used to compare the performance of the models: Adjusted R-squared, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and Root Mean Square Error (RMSE).

####Based on the table below, it is clear that the cyclical model is the best of all the models. The reason is that the cyclical model has the largest Adjusted R-squared of all the models, which means that it can best explain the data. Moreover, the cyclical model has the lowest AIC and BIC of all the models, which means that it has the best balance between fit and complexity. In addition, the cyclical model has the lowest RMSE of all the models, which means that it has the fewest errors in predictions of all the models.

####The quadratic model is superior to the linear model, because it can fit the non-linear trend of the data more effectively. However, it is not as precise as the cyclical model. The cosine model is the worst of all the models because it cannot fit the overall trend of the data, but it can fit the oscillatory pattern of the data.

####Based on the above findings, it can be concluded that the best model to use in this time series is the cyclical model i.e.the cyclical model because it can fit the data best and it is also more accurate.

```r
# Model comparison table
modelNames <- c("Linear", "Quadratic", "cyclical", "Cosine")
adjR2 <- c(summary(model1)$adj.r.squared,
           summary(model2)$adj.r.squared,
           summary(model3)$adj.r.squared,
           summary(model4)$adj.r.squared)

AICValues <- c(AIC(model1), AIC(model2), AIC(model3), AIC(model4))
BICValues <- c(BIC(model1), BIC(model2), BIC(model3), BIC(model4))

RMSEValues <- c(sqrt(mean(residuals(model1)^2)),
                sqrt(mean(residuals(model2)^2)),
                sqrt(mean(residuals(model3)^2)),
                sqrt(mean(residuals(model4)^2)))

comparisonTable <- data.frame(Model = modelNames,
                              Adjusted_R2 = adjR2,
                              AIC = AICValues,
                              BIC = BICValues,
                              RMSE = RMSEValues)

comparisonTable
```

```
##       Model Adjusted_R2      AIC      BIC     RMSE
## 1    Linear   0.6844628 1406.194 1415.265 24.21476
## 2 Quadratic   0.8045343 1334.386 1346.481 18.99493
## 3  cyclical   0.8266702 1317.093 1332.213 17.82693
## 4    Cosine   0.6591980 1418.885 1430.980 25.08149
```

Diagnostic checking:

###Plotting the standardized residuals over time to check whether the linear model assumptions are reasonable:

####The standardized residual plot displays how residuals behave over time to check if the assumptions of the linear model have been satisfied.

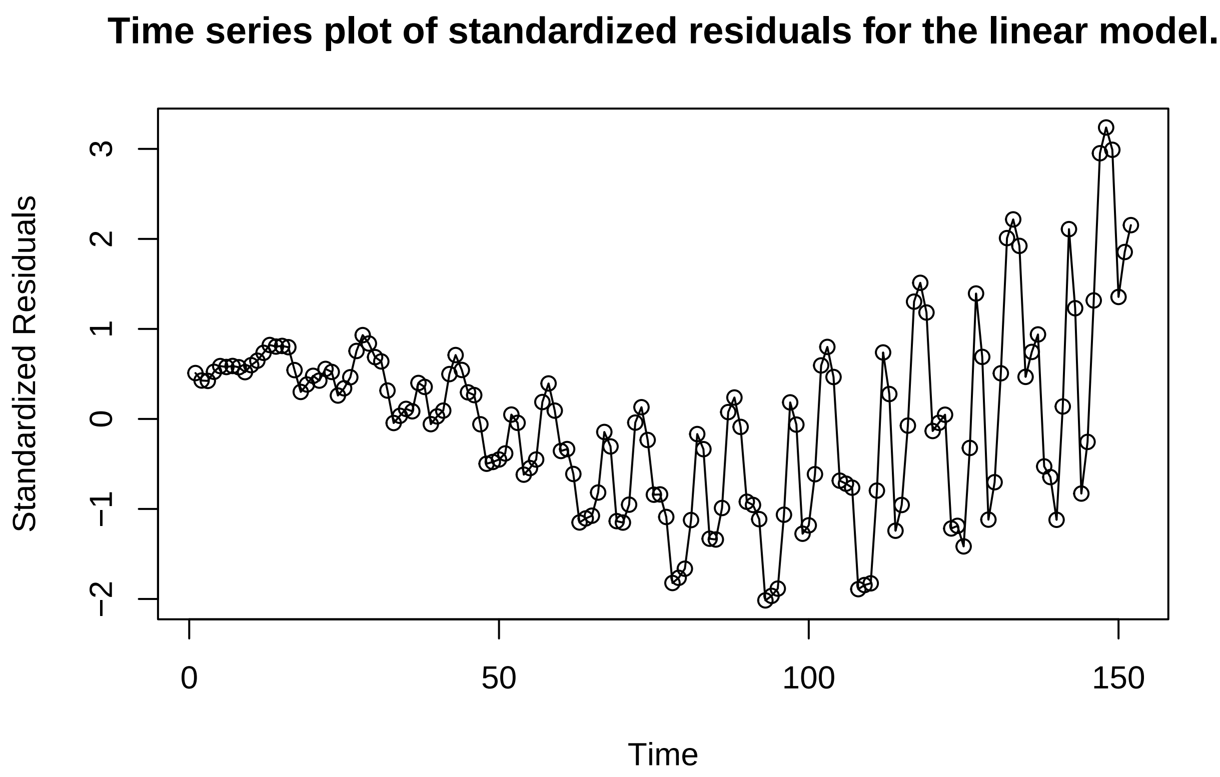
####The residuals do not behave randomly around zero and instead show a systematic behavior. This indicates that the model has not been able to capture the trend of the data.

####A structure and dependency in the residuals are evident. This is due to the fact that consecutive residuals behave in a similar manner.

####In addition to this, there is an increase in the spread of residuals over time. This increase is evident in later time periods.

####The output indicates that the assumptions of independence and constant variance have been violated. This indicates that the linear model is not suitable for this time series.

```
# Residual time series plots
# Plotting the standardized residuals over time to check whether the linear model assumptions are reaso

plot(y = rstudent(model1), x = as.vector(time(demandTS)), xlab = "Time",
     ylab = "Standardized Residuals", type = "o",
     main = "Time series plot of standardized residuals for the linear model.")
```



###Time series plot of standardized residuals for the quadratic model ####The standardized residuals plot shows the variation of the residuals with time.It is clear that the residuals are nearer to zero than the linear model. This implies that the quadratic model fits the set of data better.

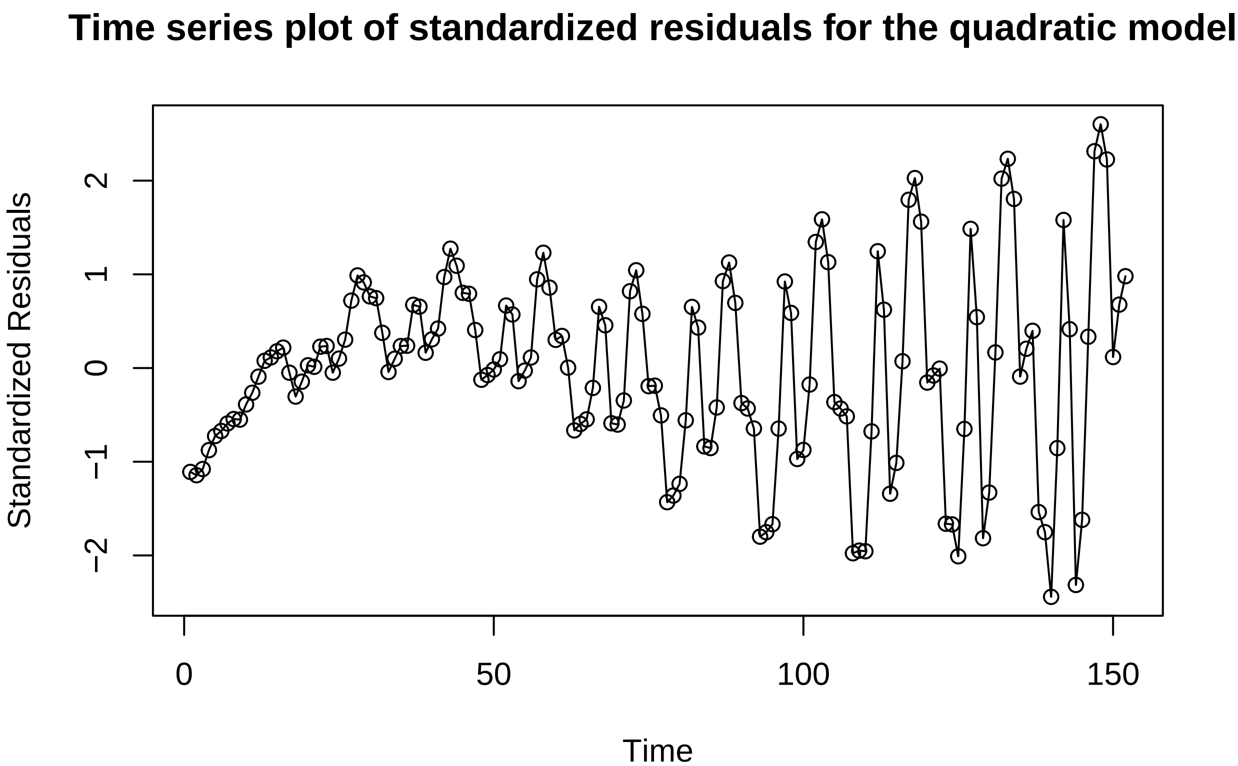
####Nonetheless, a residual pattern can still be traced. This means that the model has failed to fully remove the trends in the specified set of data.

####The dependency of the residuals on each other is very strong. This implies that the model has autocorrelation.

####Besides this, the variance of the residuals increases with time, particularly in the later periods. This means that the variance of the residuals is not fixed.

####Based on the above-presented arguments, it can be concluded that, despite the fact that the quadratic model fits the set of data provided better, its assumptions have not been met.

```
# Plotting the standardized residuals over time to check the adequacy of the quadratic model.

plot(y = rstudent(model2), x = as.vector(time(demandTS)), xlab = "Time",ylab = "Standardized Residuals"
     main = "Time series plot of standardized residuals for the quadratic model.")
```



###Time series plot of standardized residuals for the cyclical model ####The standardized residual plot is used to determine the behaviour of the residuals over time to determine whether the assumptions of the cyclical model are met.

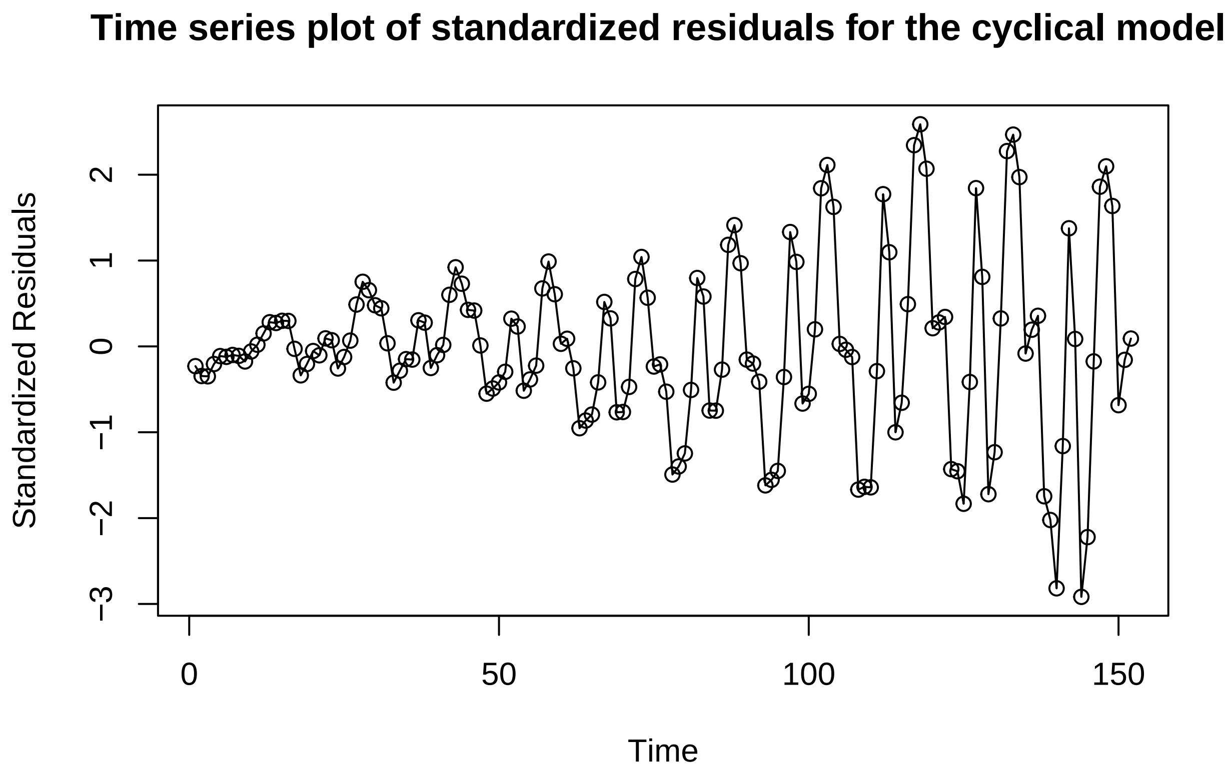
####The residuals are more randomly distributed around zero than the linear and quadratic models, which means that the cyclical model has successfully captured the majority of the underlying structure in the data.

####Pattern is less evident and the dependence between successive residuals is less, indicating that autocorrelation has been greatly reduced.

####Nevertheless, the dispersion of residuals continues to grow marginally in the later periods, which suggests that there is a certain amount of varying variance (heteroscedasticity).

####In general, the results indicate that the cyclical model is a more appropriate fit, and the independence assumption is satisfied better, but there are still slight violations of the constant variance.

```
# Plotting the standardized residuals over time to check whether the cyclical model provides an adequat

plot(y = rstudent(model3), x = as.vector(time(demandTS)), xlab = "Time",ylab = "Standardized Residuals"
     main = "Time series plot of standardized residuals for the cyclical model.")
```



###Time series plot of standardized residuals for the cosine model

####The standardized residual plot is applied to analyse the behaviour of the residuals as time goes by and gauge the validity of assumptions of the cosine model.

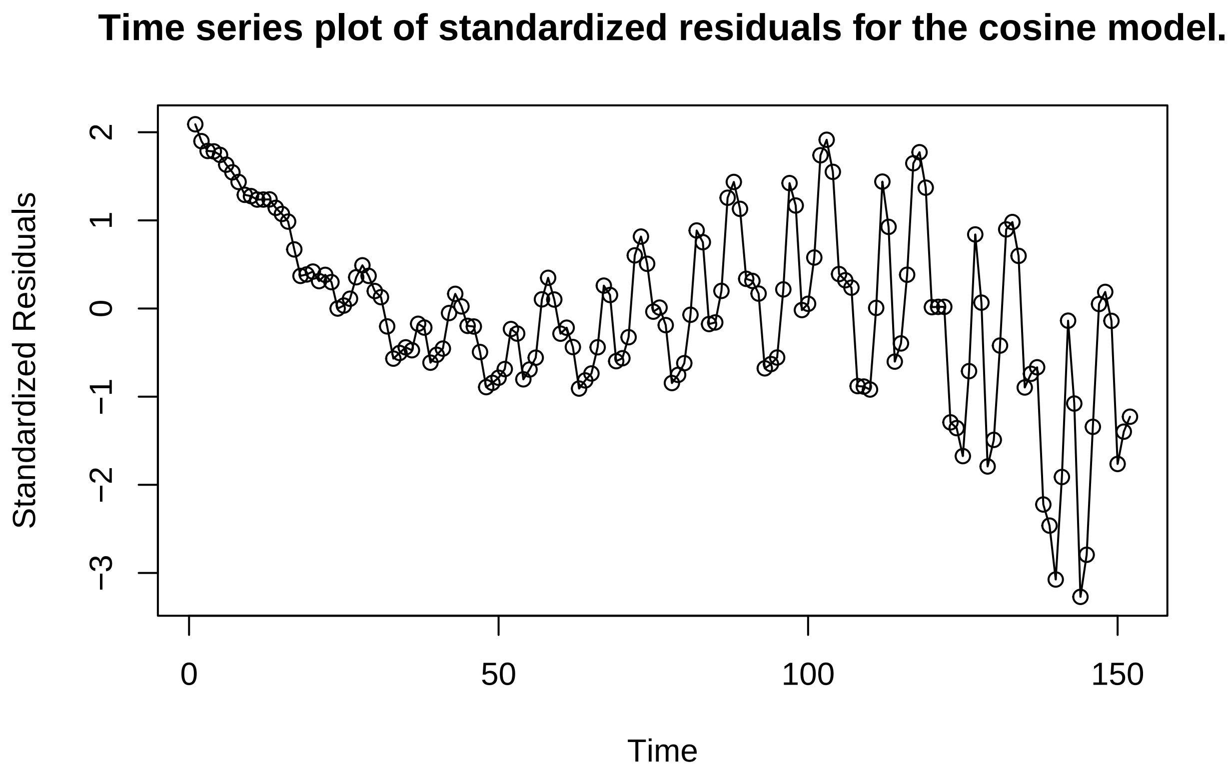
####The residuals also do not seem to be distributed randomly around the value of zero but instead have a distinct temporal trend. This suggests that the cosine model has not been able to take into account the latent structure of the data.

####A clear cyclic trend is observed in the residuals indicating that the observations have autocorrelation and dependence.

####Moreover, the dispersion of the residuals becomes somewhat higher in the later time intervals, which means that the heteroscedasticity also exists to some extent.

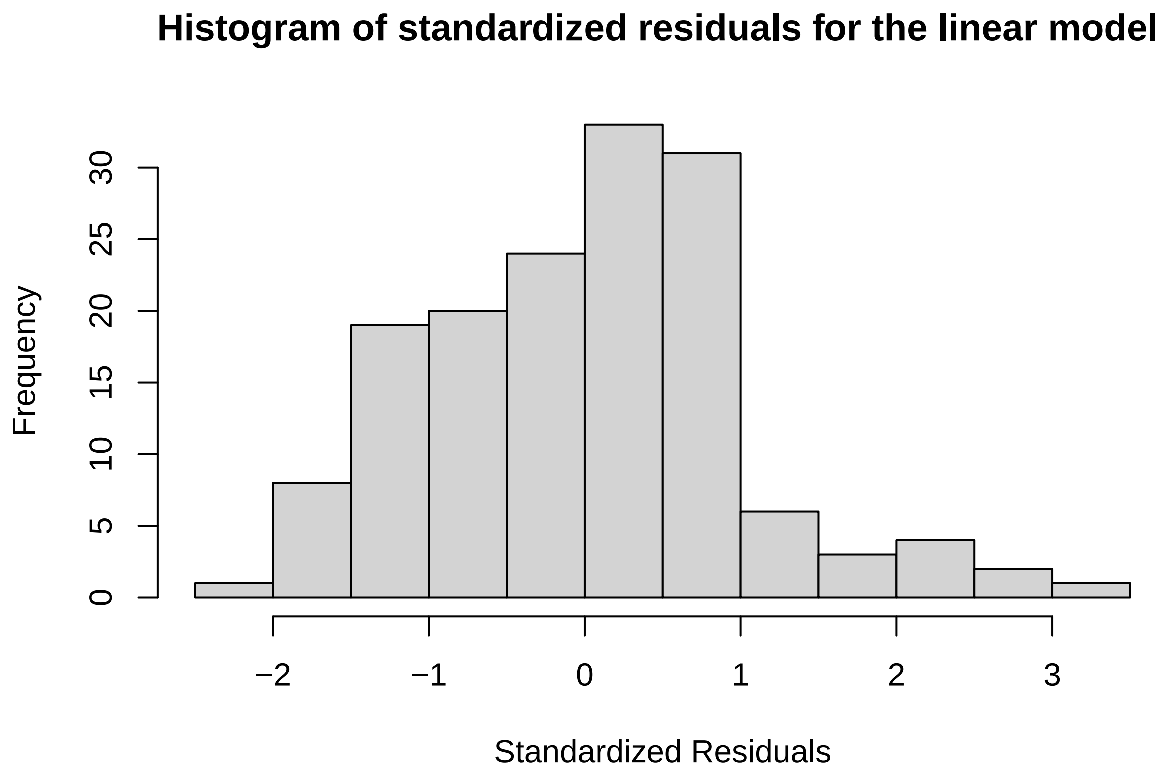
####In general, it is implied in the plot that the cosine model does not provide sufficient fit to the data because the assumptions of the independence and constant variance are not completely met.

```
# Plotting the standardized residuals over time to assess whether the cosine model fits the series adeq

plot(y = rstudent(model4), x = as.vector(time(demandTS)), xlab = "Time", ylab = "Standardized Residuals
```



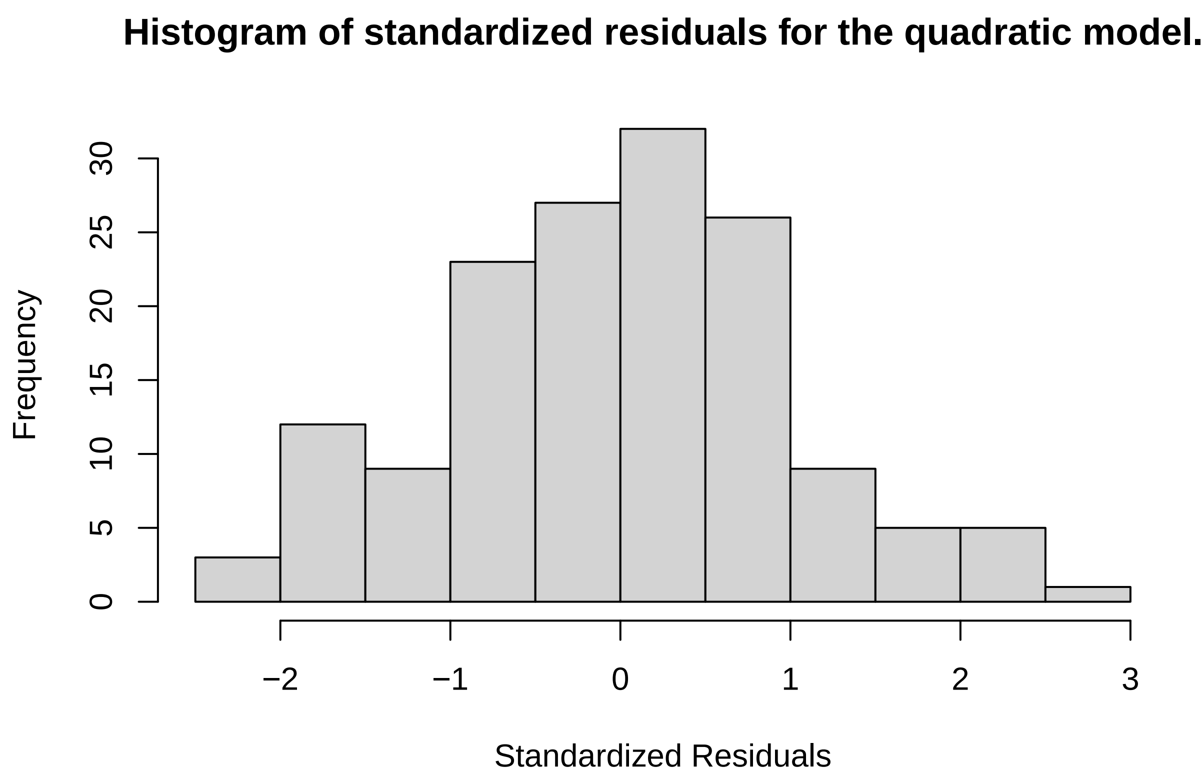
###Histogram of standardized residuals for the linear model ####The output shows the distribution of standardized residuals of the linear model. It is evident that the majority of the residuals are clustered around zero. This implies that the model errors are concentrated. But it is also evident that the distribution is not symmetric and is not a perfect bell shape. It is more distributed on one side and also has extreme values. This implies that the residuals are not normally distributed. This, in its turn, implies that the linear model does not satisfy the normality assumption.

```r
# Histograms
# Displaying the distribution of standardized residuals to assess normality for the linear model.
hist(rstudent(model1), xlab = "Standardized Residuals",main = "Histogram of standardized residuals for t
```



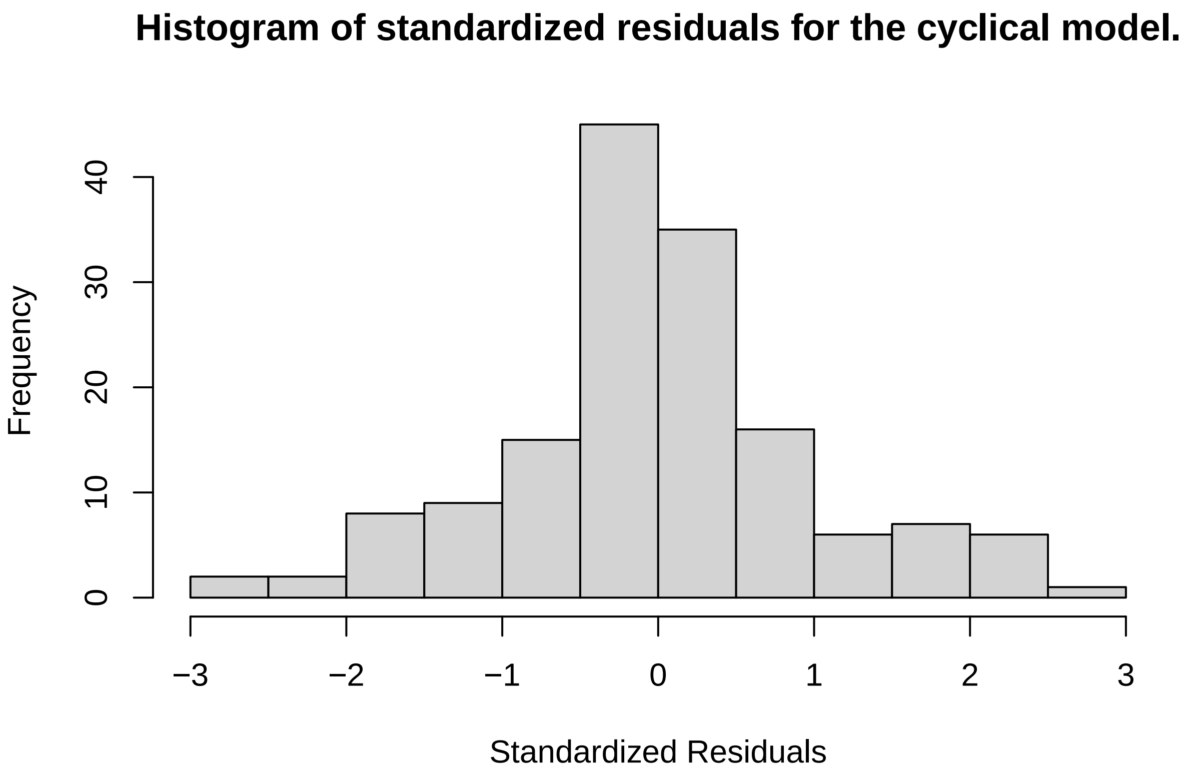
###Histogram of standardized residuals for the quadratic model ####The output displays the distribution of standardized residuals of the quadratic model. The majority of the residuals are clustered around zero and the distribution is more symmetric than the linear model. It is more of a bell-shaped curve, but there are still some values in the tails. This implies that the residuals are normally distributed, implying that the normality assumption is fairly met in the quadratic model.

```
# Displaying the distribution of standardized residuals to assess normality for the quadratic model.
hist(rstudent(model2), xlab = "Standardized Residuals",main = "Histogram of standardized residuals for t
```



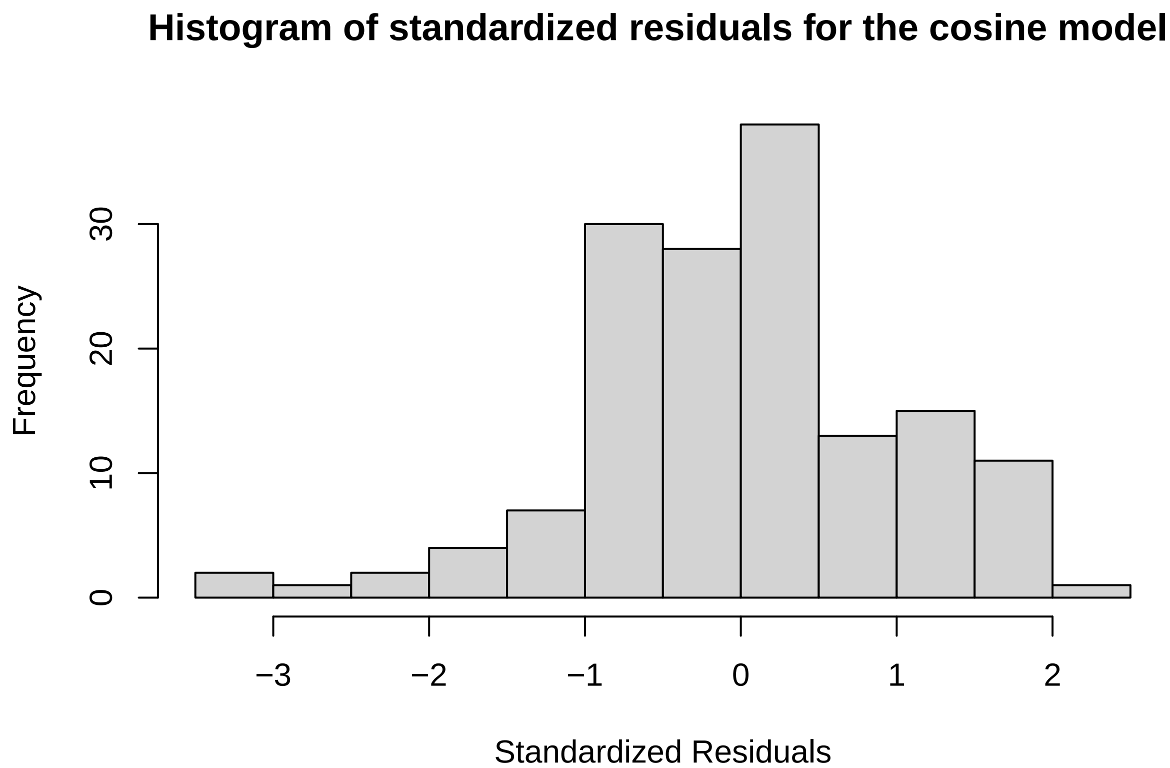
###Histogram of standardized residuals for the cyclical model ####The output indicates the distribution of standardized residuals of the cyclical model. The distribution is quite symmetric and most of the residuals are concentrated around zero. The form is similar to a bell-shaped curve, where the values are more concentrated in the center and less in the tails. There are some extreme values, but they are not numerous. This implies that the residuals are normally distributed, implying that the normality assumption is fairly met in the cyclical model.

```
# Displaying the distribution of standardized residuals to assess normality for the cyclical model.
hist(rstudent(model3), xlab = "Standardized Residuals",
     main = "Histogram of standardized residuals for the cyclical model.")
```



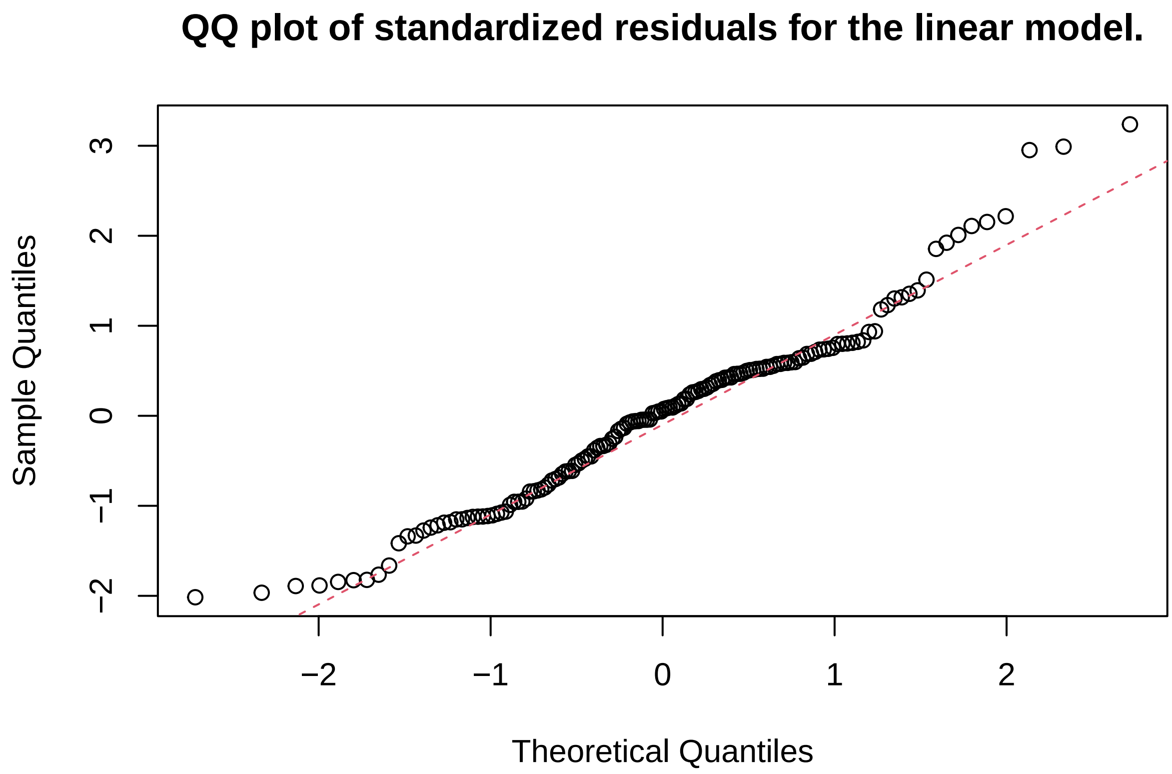
###Histogram of standardized residuals for the cosine model ####The output indicates the distribution of the standardized residuals for the cosine model. The residuals are not perfectly symmetric, and their distribution is not perfectly bell-shaped. It can be observed that there is a significant degree of skewness and increased variability, or "spread," in the distribution of residuals, especially on one side. This suggests that the residuals are not normally distributed, and hence, it can be inferred that the normality condition is not satisfied for this model.

```r
# Displaying the distribution of standardized residuals to assess normality for the cosine model.
hist(rstudent(model4), xlab = "Standardized Residuals",
     main = "Histogram of standardized residuals for the cosine model.")
```



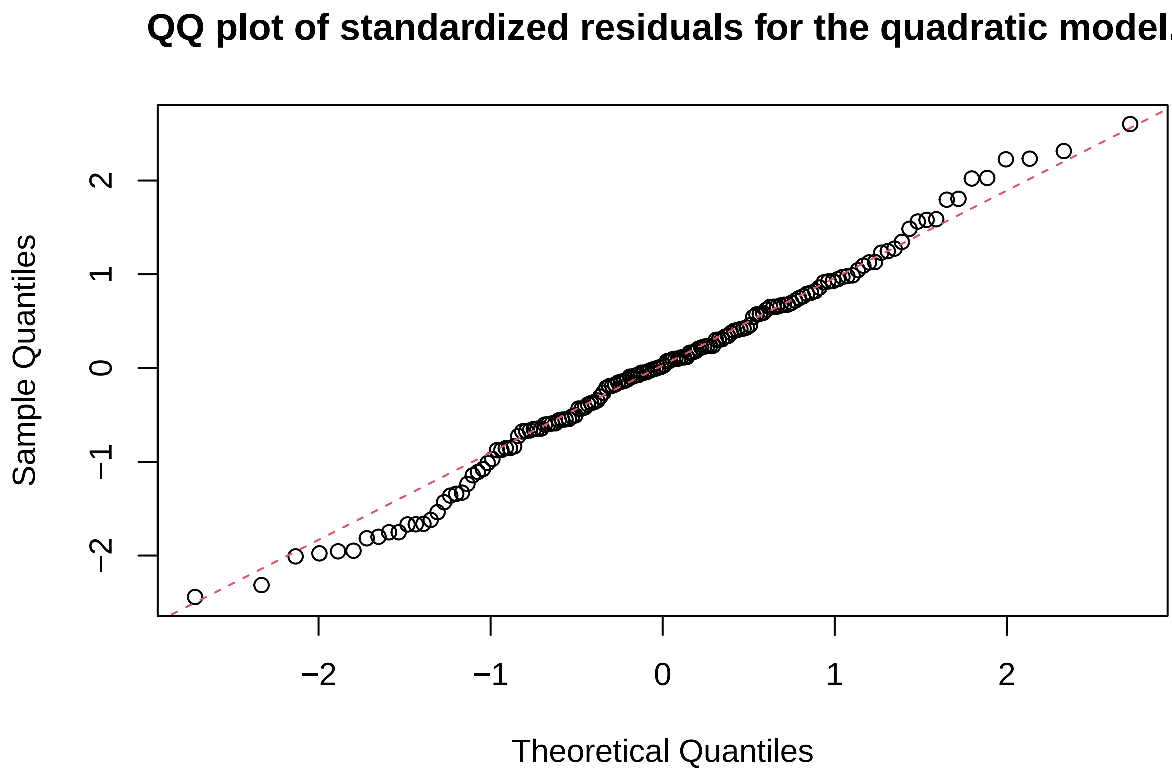
###QQ plots: ###QQ plot of standardized residuals for the linear model ####I observed that the majority of the points are close to the reference line. This implies that the residuals are normally distributed. However, I also observed where the points move away from the line at the tails. This means that the points are not normally distributed. Even though the points are normally distributed around the center point, they are not normally distributed around the extremes.

```
# Using a QQ plot to check if the standardized residuals from the linear model are normally distributed
y <- rstudent(model1)
qqnorm(y, main = "QQ plot of standardized residuals for the linear model.")
qqline(y, col = 2, lwd = 1, lty = 2)
```



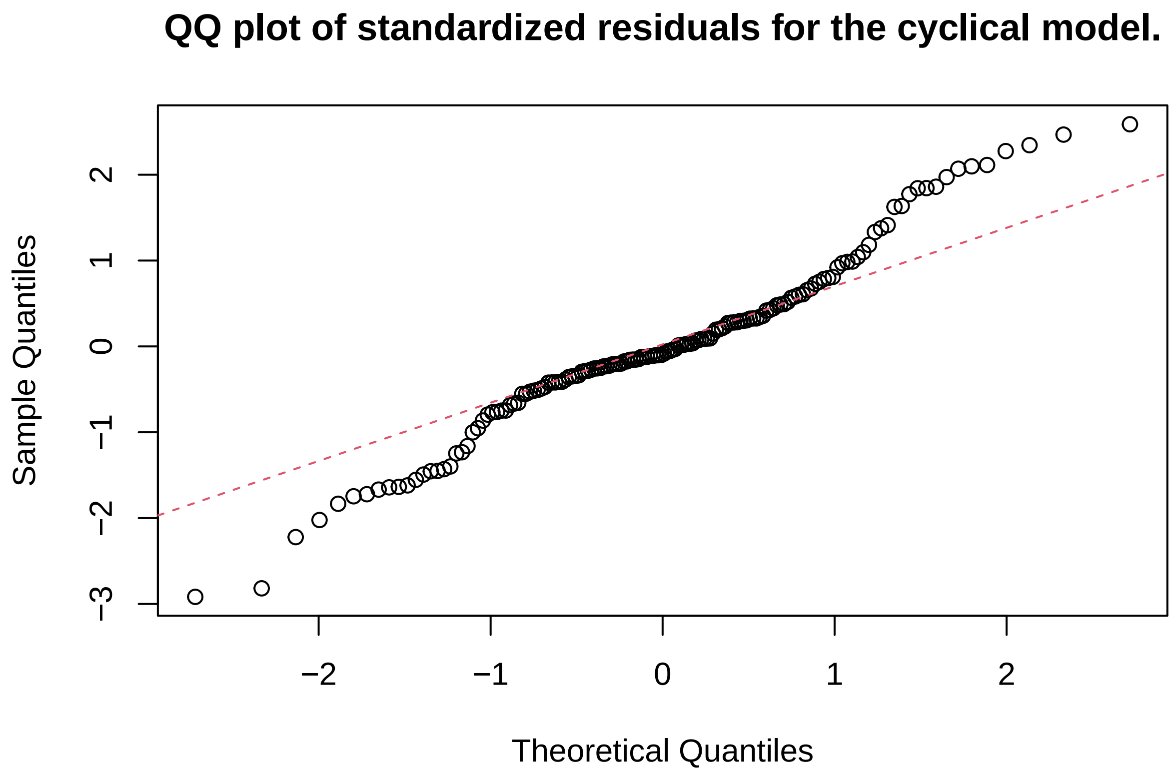
###QQ plot of standardized residuals for the quadratic model ####The majority of the points are very near the reference line, which means that the residuals are normally distributed. The alignment is stronger than in the linear model, especially in the central part, with only a few points slightly out of the line at the tails, which implies that the normality assumption is more met in the quadratic model, and the model fits better with fewer normality violations.

```
# Using a QQ plot to check whether the standardized residuals from the quadratic model are approximatel
y <- rstudent(model2)
qqnorm(y, main = "QQ plot of standardized residuals for the quadratic model.")
qqline(y, col = 2, lwd = 1, lty = 2)
```



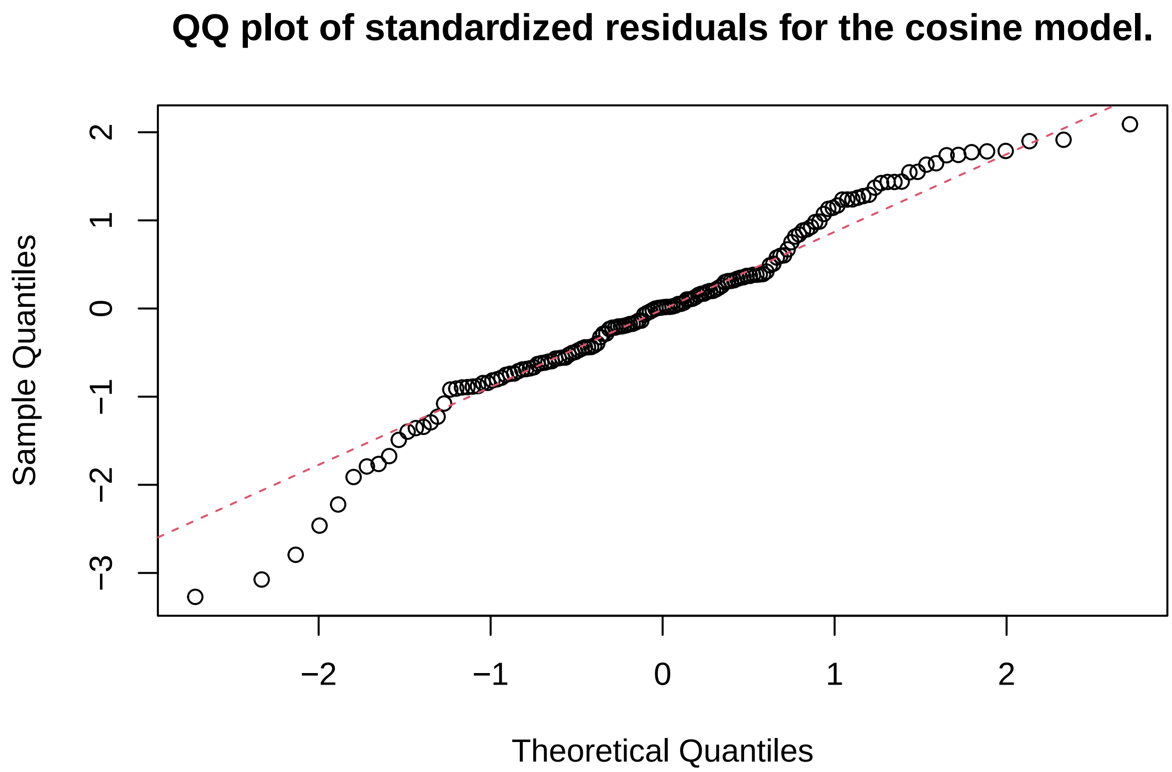
###QQ plot of standardized residuals for the cyclical model ####Most of these points lie close to the line. It also indicates that the residuals tend to be normally distributed. However, there are signs of variation on both ends of the curve. More significantly, there is variation on the lower end of the curve, with points significantly below the line. The model appears to be performing well around the middle. However, there is more variation than was seen with the quadratic model. This indicates that the cyclical model is not entirely normally distributed and there is evidence of heavier tails and possibly outliers.

```
# Using a QQ plot to check whether the standardized residuals from the cyclical model are approximately
y <- rstudent(model3)
qqnorm(y, main = "QQ plot of standardized residuals for the cyclical model.")
qqline(y, col = 2, lwd = 1, lty = 2)
```



###QQ plot of standardized residuals for the cosine model ####Most of the points are near the reference line in the central region, suggesting approximate normality in this region.However, significant deviations are evident at both ends, especially at the lower end, where points are substantially below the line.The significant deviations at both ends of the plot suggest that the residuals are not normally distributed. The assumption of normality is not well satisfied for the cosine model.

```
# Using a QQ plot to check whether the standardized residuals from the cosine model are approximately n
y <- rstudent(model4)
qqnorm(y, main = "QQ plot of standardized residuals for the cosine model.")
qqline(y, col = 2, lwd = 1, lty = 2)
```



###Shapiro-Wilk tests(linear model) ####In the case of the linear model, the p-value is 0.003394, which is not greater than 0.05. This implies that the null hypothesis is rejected, which implies that the residuals are not normally distributed, thereby proving that the normality condition is not satisfied for the linear model.

```
# Shapiro-Wilk tests:
# Using the Shapiro-Wilk test to check normality of the linear model residuals.
y <- rstudent(model1)
shapiro.test(y)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  y
## W = 0.97198, p-value = 0.003394
```

###Shapiro-Wilk tests(Quadratic model)

####In the quadratic model, the p-value for the Shapiro-Wilk test is 0.4466, which does not fall in the range less than 0.05. Therefore, the null hypothesis of normality cannot be rejected, and the normality of the residuals can be concluded, which proves that the normality assumption holds for the quadratic model.

```
# Using the Shapiro-Wilk test to check normality of the quadratic model residuals.
y <- rstudent(model2)
shapiro.test(y)
```

```
##
```

```
##  Shapiro-Wilk normality test
##
## data:  y
## W = 0.99098, p-value = 0.4466
```

###Shapiro-Wilk tests(cyclical model) ####The Shapiro Wilk test is used for testing whether or not the standardized residuals of the cyclical model are normally distributed, or if it can be assumed that they are, or not.The value of p, in this case, is 0.003402, which is lower than the significance of 0.05. This means that the null hypothesis, which states that the data is normally distributed, is rejected.This indicates that the residues of the cyclical model are not normally distributed. This, in turn, indicates that the normality condition is not being met.However, even though this is an infraction, it can be noted that the residuals are more centered at zero and are more random than those of other models. This is an indication that the nature of the data is better represented by the cyclical model, even though it is not perfectly normal.

```
# Using the Shapiro-Wilk test to check normality of the cyclical model residuals.
y <- rstudent(model3)
shapiro.test(y)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  y
## W = 0.97199, p-value = 0.003402
```

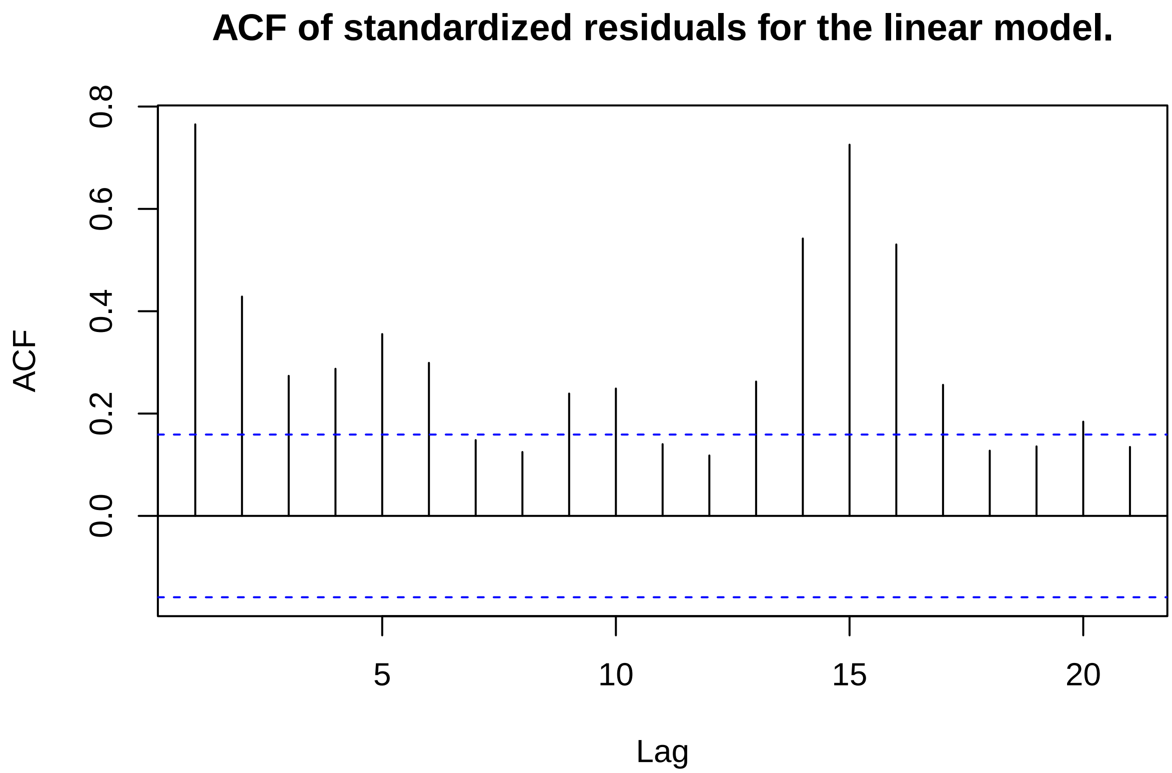
###Shapiro-Wilk tests(cosine model) ####For the cosine model, the Shapiro-Wilk test gives a p-value of 0.006071, which is less than 0.05. This means the residuals are not normally distributed, indicating that the normality assumption is not satisfied for this model.

```
# Using the Shapiro-Wilk test to check normality of the cosine model residuals.
y <- rstudent(model4)
shapiro.test(y)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  y
## W = 0.97433, p-value = 0.006071
```

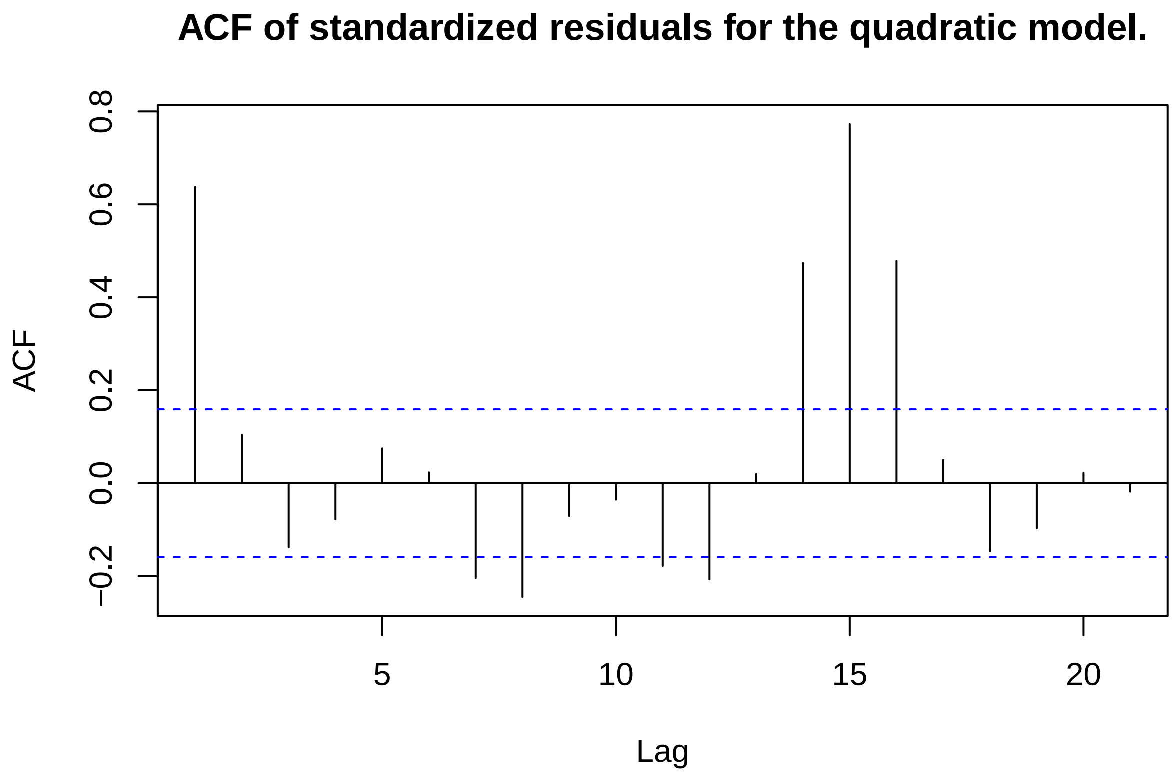
###ACF of standardized residuals for the linear model ####In the linear model ACF plot, there are a few spikes exceeding the significance levels, particularly at low and a few high lags. This shows that the residuals have not become random over time. These great autocorrelations indicate that these models have not fitted all the trends in the data. In general, this implies that the assumption of independence does not hold and the linear model does not fit this time series.

```
# ACF plots of standardized residuals
# Using the ACF plot to check autocorrelation in the linear model residuals.
acf(rstudent(model1), main = "ACF of standardized residuals for the linear model.")
```



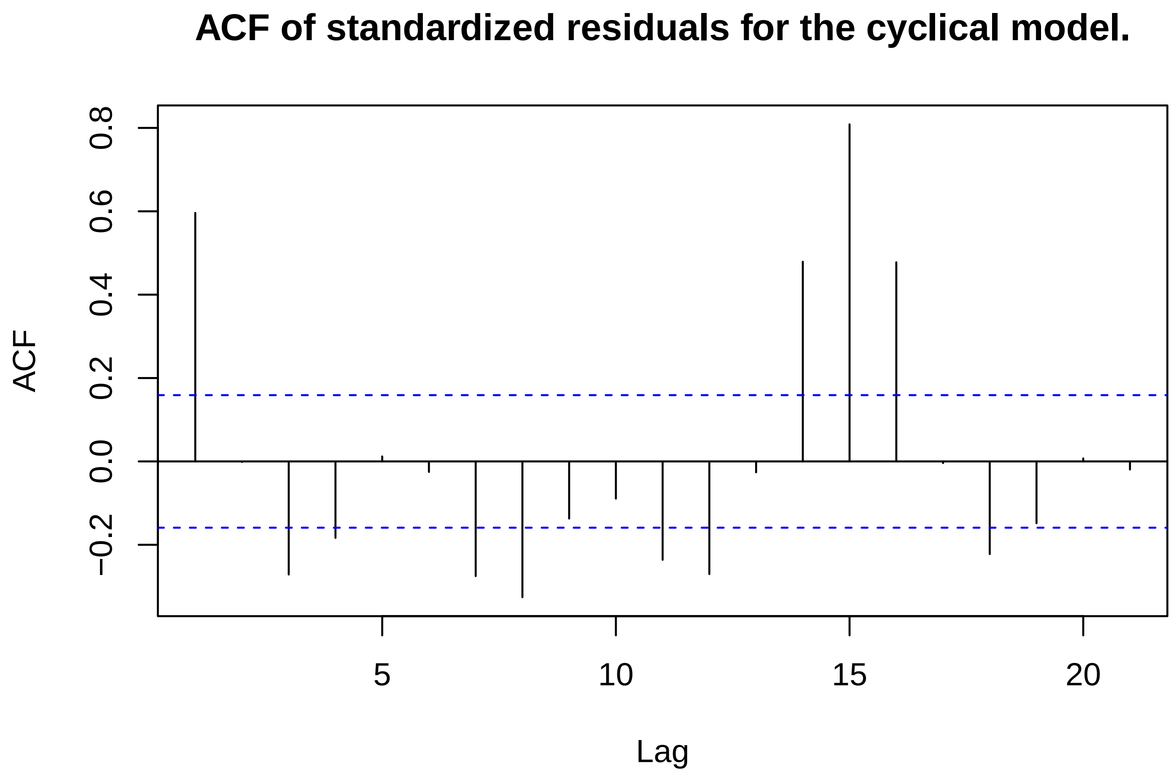
###ACF of standardized residuals for the quadratic model ####In the case of the quadratic model, the ACF plot of the model has fewer spikes with significance values being beyond the significance limits than what have been observed with the linear model, implying that there is less autocorrelation in the quadratic model than in the linear model. Nevertheless, even some spikes, particularly at some lags (such as the mid-range lags), are over the boundaries indicating that not all the residuals are independent. This is to say that the quadratic model will provide a better fit but it still contains some residual autocorrelation, and the assumption of independence is not met fully.

```
# Using the ACF plot to check autocorrelation in the quadratic model residuals.
acf(rstudent(model2), main = "ACF of standardized residuals for the quadratic model.")
```



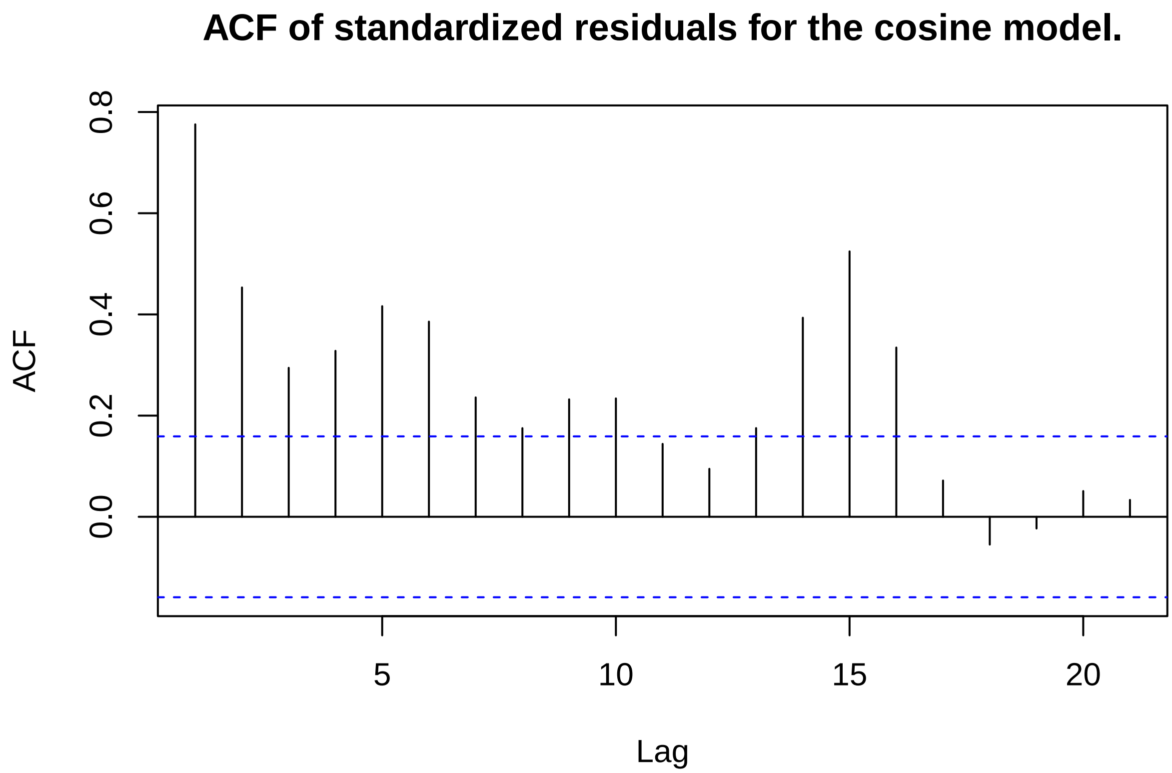
###ACF of standardized residuals for the cyclical model ####The ACF plot, in the cyclical model, still has a few larger spikes, beyond the significance levels, at some lower and intermediate lags. This does not mean that the residuals do not have autocorrelation anymore. Although the cyclical model is more structured than the more basic models, the residual is not a completely random one. This way, it is not satisfied in the assumption of independence, meaning that the model cannot explain all of the trends in the data.

```
# Using the ACF plot to check autocorrelation in the cyclical model residuals.
acf(rstudent(model3), main = "ACF of standardized residuals for the cyclical model.")
```



###ACF of standardized residuals for the cosine model ####In the case of the cosine model, ACF plot indicates that there are still a number of spikes that have surpassed the significance limit especially at lower lags and some of the mid-range lags. This also implies that the residuals still have autocorrelation. Even though the cosine model helps to explain certain cyclical behaviour, the residual is not entirely random. Thus the independence assumption is not completely met, which implies that the model is not a complete reflection of all the underlying patterns in the data.

```r
# Using the ACF plot to check autocorrelation in the cosine model residuals.
acf(rstudent(model4), main = "ACF of standardized residuals for the cosine model.")
```



###Forecast Results and Interpretation (Cyclical Model) ####The cyclical model produces forecasts in the coming 15 time periods, as well as lower and upper prediction limits. The result of the output shows that the values of forecast (fit) rise in 70.12 in step 1 to 103.30 in step 15 with a continuous increase in the predicted energy demand over time. The lower bounds (lwr) move upwards 32.50 to 61.13, and the upper bounds (upr) move upwards 107.75 to 145.46 indicating that the overall prediction interval moves as well as forecasts. It is also noted that the prediction intervals grow slowly, the difference between upper and lower limits of prediction interval increases with time. This is indicative of growing doubt on long-term predictions. In general, the findings indicate that the cyclical model forecasts an increasing demand trend, though with an increasing variation, and therefore, short-term forecasts are better than long-term forecasts.

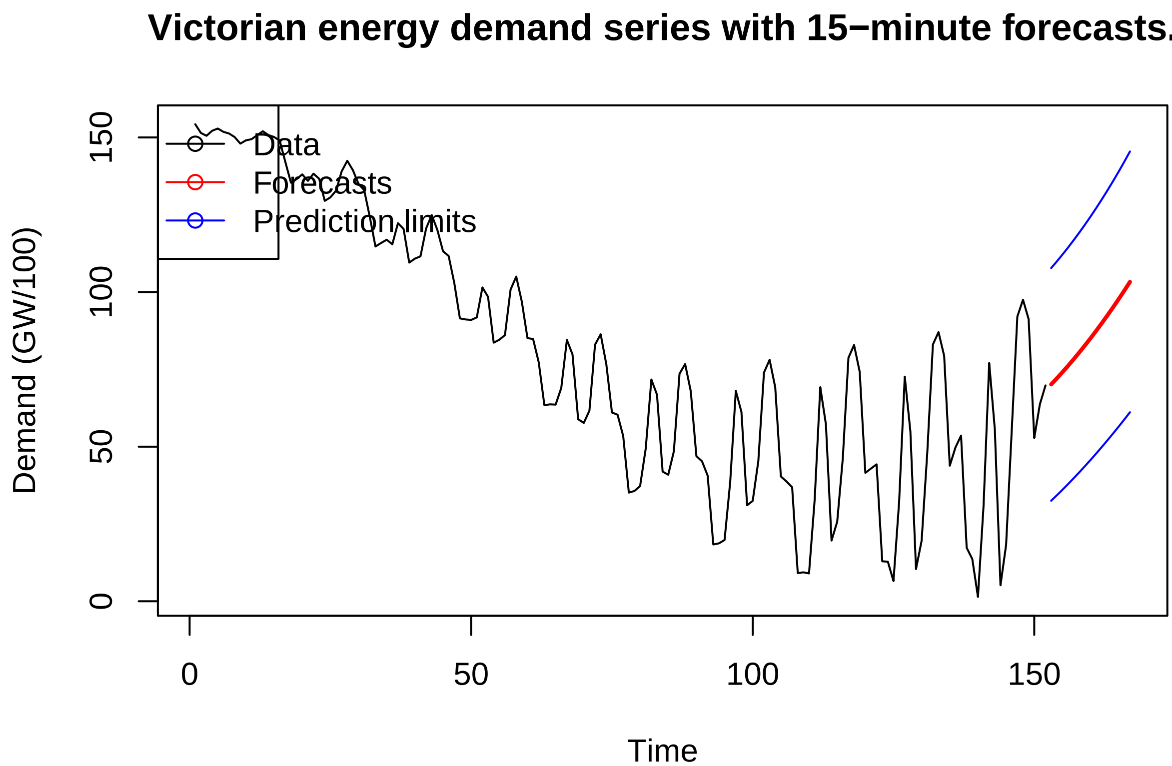
```
# Forecasting on cyclical model
#Using the best fitted model, the cyclical model, to generate forecasts for 15 steps ahead and provide 


h <- 15

future_t <- seq(length(t) + 1, length(t) + h, 1)

aheadTimes <- data.frame(t = future_t)

forecasts <- predict(model3, newdata = aheadTimes, interval = "prediction")
print(forecasts)
```

```
##          fit      lwr      upr
## 1   70.12392 32.49977 107.7481
## 2   72.06600 34.24948 109.8825
```

```
## 3   74.07149 36.04585 112.0971
## 4   76.14098 37.88859 114.3934
## 5   78.27506 39.77739 116.7727
## 6   80.47433 41.71195 119.2367
## 7   82.73938 43.69196 121.7868
## 8   85.07080 45.71711 124.4245
## 9   87.46918 47.78710 127.1513
## 10  89.93513 49.90163 129.9686
## 11  92.46922 52.06043 132.8780
## 12  95.07206 54.26322 135.8809
## 13  97.74424 56.50976 138.9787
## 14 100.48634 58.79981 142.1729
## 15 103.29897 61.13318 145.4648
```

###Plotting the data with forecasts and prediction limits from the cyclical model ####Output presents the original time series and the forecasted value and prediction intervals. The black line (Data) is the real values and the red line (Forecasts) is the forecasted values of the 15 time periods ahead. The projections are evidently in a progressive trend which means that the demand is likely to grow. The blue lines (Prediction limits) signify the lower and upper limit of the forecasts, and both limits too change upwards and this remains in line with the trend that is being predicted. It is also noted that the distance existing between the upper and lower limits grows as time goes and this means that the uncertainty increases with the increase in the forecast horizon. In general, it indicates that the cyclical model is an adequate description of data and reasonable predictions; however, longer-term predictions are accompanied by more uncertainty.

```r
# Plotting the data with forecasts and prediction limits from the cyclical model.

plot(demandTS, xlim = c(1, length(demandTS) + h),
     ylim = c(min(c(demandTS, forecasts)), max(c(demandTS, forecasts))),
     ylab = "Demand (GW/100)",
     main = "Victorian energy demand series with 15-minute forecasts.")

lines(ts(as.vector(forecasts[,1]), start = length(demandTS) + 1, frequency = 1),
      col = "red", type = "l", lwd = 2)

lines(ts(as.vector(forecasts[,2]), start = length(demandTS) + 1, frequency = 1),
      col = "blue", type = "l", lwd = 1)

lines(ts(as.vector(forecasts[,3]), start = length(demandTS) + 1, frequency = 1),
      col = "blue", type = "l", lwd = 1)

legend("topleft", lty = 1, pch = 1, col = c("black", "red", "blue"),
       text.width = 2,
       c("Data", "Forecasts", "Prediction limits"))
```



###Conclusion: ####Finally, the current work has examined the Victorian energy demand time series with the help of various regression-based models, such as linear, quadratic, cyclical (cubic), and cosine, with the view to determining the most suitable representation of the underlying data structure. The comparative analysis based on Adjusted R^2, AIC, BIC, and RMSE showed that the cyclical model was the most appropriate as it gives the most optimum balance between good fit and complexity schemes and includes the non-linear behaviour exhibited by the series.

####This choice is substantiated by the diagnostic analysis in that the residuals of the cyclical model appear to follow a more random distribution around zero with less autocorrelation than the other models. Nevertheless, there are still small violations of model assumptions, especially small amounts of heteroscedasticity and residual autocorrelation, which means that although the model has a significant positive effect on the fit, it does not completely eliminate all the underlying interdependence of data.

####The results of the prediction using the cyclical model show that the future energy demand will show gradual positive growth tendency, but the prediction interval will grow as time passes, as the predictability will decrease as the forecast horizon goes further. On the whole, the cyclical model is quite reliable and interpretable and thus is regarded as the most suitable model in the short-term forecasting in this analysis.